A Project Report on

LSTM Autoencoder-Based Persistent and Transient Anomaly Classification in Open RAN KPIs

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

in

A.Y.: 2026-27

by

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DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Name of Title: LSTM Autoencoder-Based Persistent and Transient Anomaly Classification in Open RAN KPIs

1. Abstract:

The increasing complexity of modern 5G and Open Radio Access Networks (Open RAN) makes continuous monitoring of network performance essential for maintaining reliable and efficient communication services. Open RAN systems generate large volumes of Key Performance Indicator (KPI) data representing different aspects of network and radio performance. Abnormal changes in these KPIs may occur because of temporary traffic variations, interference, resource congestion, equipment degradation, configuration problems, or other network conditions. A major challenge for network operators is that an abnormal KPI value does not always indicate a fault requiring immediate corrective action. A temporary deviation may disappear within a short period, while a persistent deviation may indicate a continuing network problem. Conventional threshold-based monitoring can detect deviations but may not effectively capture the temporal behaviour of these events, which can lead to unnecessary alarms and difficulty in prioritizing important faults. This project proposes an LSTM (Long Short-Term Memory) Autoencoder-based deep-learning approach for detecting and classifying anomalies in Open RAN KPI time-series data. The collected KPI data will first undergo preprocessing, including timestamp handling, missing-value treatment, feature selection, scaling, and time-window formation. The LSTM Autoencoder will be trained primarily using normal KPI sequences so that it can learn the temporal characteristics of healthy network operation. The model will reconstruct the input KPI sequences, and the reconstruction error between the original and reconstructed sequences will be used to identify abnormal behaviour. A suitable anomaly threshold will then be determined using the training and validation data rather than relying only on an arbitrary fixed value.

Once an anomalous sequence is detected, the proposed system will perform temporal analysis based on the duration, magnitude, continuity, and recovery behaviour of the detected abnormality to distinguish transient anomalies from persistent anomalies. A transient anomaly represents a short-term deviation that returns to normal behaviour, whereas a persistent anomaly continues for a longer period and may require investigation or corrective action by the network operator. Clustering techniques may also be explored to group similar anomalous patterns and identify recurring network behaviours. The complete solution will follow the machine-learning lifecycle of data preprocessing, feature preparation, LSTM Autoencoder training, validation, anomaly detection, classification, and performance evaluation. The proposed approach will be evaluated using metrics such as precision, recall, F1-score, accuracy where applicable, ROC-AUC, false-positive rate, reconstruction error, and detection delay. For transient and persistent classification, classification accuracy and F1-score will be considered along with the rate of false alarms, since incorrectly treating a temporary event as a persistent fault can lead to unnecessary operator intervention. The results will also be compared with simpler anomaly-

detection approaches where appropriate to determine the effectiveness of the LSTM-based method. The primary objective is not only to detect abnormal KPI behaviour but also to understand its temporal nature and provide more meaningful information to network operators. By distinguishing short-lived disturbances from persistent degradation, the proposed system can help operators prioritize anomalies that are more likely to require corrective action while reducing unnecessary responses to temporary events. This deep-learning-based framework therefore provides a practical foundation for intelligent Open RAN KPI monitoring and further research into early identification of persistent network problems.

2. SOFTWARE REQUIREMENTS

a) **Operating System:** Windows 10/11
b) **Programming Language:** Python 3.10 or later
c) **Development Environment:** Visual Studio Code / Jupyter Notebook
d) **Data Processing:** Pandas, NumPy
e) **Data Visualization:** Matplotlib, Seaborn
f) **Machine Learning:** Scikit-learn
g) **Deep Learning:** TensorFlow / Keras
h) **Dataset:** Open RAN KPI time-series dataset in CSV format
i) **Model:** LSTM Autoencoder
j) **Supporting Libraries:** SciPy

Signature of the Supervisor Course Instructor